

INTERNSHIP PROJECT REPORT

Project Title

YourTube – Intelligent Video Streaming Platform with Personalized User Experience and Secure Authentication

1. Introduction

The objective of this internship project was to enhance an existing video streaming platform by implementing advanced features related to user authentication, personalization, content moderation, accessibility, and user engagement.

The project was developed as a full-stack web application using modern web technologies. Several real-world functionalities such as OTP verification, dynamic theme switching, multilingual comment translation, and automated moderation were implemented to improve both usability and security.

2. Technologies Used

Frontend Technologies

- Next.js
- React.js
- TypeScript
- JavaScript
- Tailwind CSS
- ShadCN UI Components
- Axios
- Firebase Authentication
- Lucide React Icons

Backend Technologies

- Node.js
- Express.js
- MongoDB
- Mongoose ODM
- Nodemailer
- Crypto Module
- Translate-Google Package

Authentication Technologies

- Firebase Google Authentication
- Email OTP Verification
- Simulated SMS OTP Service

Database

- MongoDB Atlas / MongoDB
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3. Task 1 – User Authentication System

Objective

To allow users to securely authenticate themselves before accessing platform features.

Implementation

The authentication system was implemented using Firebase Google Authentication.

Features Implemented

- Google Sign-In
- Persistent Login Sessions
- Secure Logout Functionality
- User Data Synchronization with MongoDB

Workflow

1. User clicks Google Sign-In.
2. Firebase authenticates the user.
3. User information is sent to the backend.
4. Backend checks MongoDB.
5. If user does not exist, a new user record is created.
6. Existing users are logged in directly.

Benefits

- Secure authentication
 - Fast login experience
 - Reduced password management complexity
-

4. Task 2 – Video Management and User Interaction

Objective

To allow users to interact with videos.

Features Implemented

Video Features

- Upload Videos
- View Videos
- Track Views
- Watch History

User Interaction Features

- Like Videos
- Dislike Videos
- Watch Later
- Download Videos
- Share Videos

Database Storage

MongoDB collections were used to maintain:

- Video Information
- User Activity
- View Count
- Watch Later Data
- Download History

Benefits

- Increased user engagement
 - Better content discovery
 - Personalized viewing experience
-

5. Task 3 – Premium Features and Download Management

Objective

To implement premium-related functionality and content download tracking.

Features Implemented

Download Tracking

Each downloaded video is stored with:

- Video ID
- Download Timestamp

User Plan Management

Implemented plan-based architecture:

- Free Plan
- Premium Plan (Extendable)

Benefits

- Monetization readiness
 - Download analytics
 - User subscription support
-

6. Task 4 – Profile Management System

Objective

To collect additional user information required for personalized services.

Features Implemented

User Profile Fields

- Name
- Phone Number
- State

- City
- Channel Name
- Description

Profile Completion Dialog

A mandatory profile completion dialog was developed.

Features

- Validation of required fields
- Prevents incomplete profile creation
- Automatically appears for new users

Benefits

- Improved user identification
 - Personalized feature delivery
 - Better user data management
-

7. Task 5 – Context-Aware Theme and Region-Based Authentication

Objective

To personalize user experience based on geographical location and login time while enhancing authentication security.

Part A – Dynamic Theme Switching

Requirement

If:

- State belongs to South India
- Time is between 10:00 AM and 12:00 PM IST

Then:

- Light Theme

Else:

- Dark Theme

Supported South Indian States

- Tamil Nadu
- Kerala
- Karnataka
- Andhra Pradesh
- Telangana

Implementation

The system automatically:

1. Detects user state.
2. Calculates current IST time.
3. Applies corresponding theme.

Result

Condition	Theme
South India + 10 AM-12 PM	Light
Any Other Case	Dark

Part B – Region-Based OTP Authentication

Requirement

South Indian Users:

- OTP sent via Email

Other Users:

- OTP sent via Mobile SMS

Implementation

Email OTP

Implemented using:

- Nodemailer

- Gmail SMTP

Mobile OTP

Implemented through:

- Simulated SMS Service Layer

The architecture supports future integration with:

- Twilio
- MSG91
- Fast2SMS

without modifying business logic.

Security Features Added

OTP Expiry

- 5-minute validity

OTP Cooldown

- 30-second resend restriction

OTP Attempt Limit

- Maximum 5 attempts

Verification Status

Stored in database using:

```
isVerified
```

Benefits

- Secure authentication
 - Region-based verification
 - Future scalability
-

8. Task 6 – Intelligent Comment Moderation and Translation System

Objective

To create a safer, multilingual, and more inclusive comment section.

Part A – City-Based Context Display

Each comment displays:

- Username
- City
- Timestamp

Example:

Sparsh Singh • Ajmer
2 minutes ago

Benefits

- Improved context
 - Better community engagement
-

Part B – Comment Likes and Dislikes

Features

Users can:

- Like Comments
- Dislike Comments

Protection Implemented

One User:

- One Like
- One Dislike

Duplicate voting is blocked.

Database Fields

```
likes  
dislikes  
likedBy  
dislikedBy
```

Part C – Automatic Comment Removal

Requirement

If a comment receives:

```
2 Dislikes
```

Then:

```
Comment Automatically Deleted
```

Implementation

Automatic moderation logic was added to backend controllers.

Benefits

- Self-moderated community
 - Reduced offensive content
-

Part D – Special Character Filtering

Requirement

Comments containing prohibited special characters must be blocked.

Validation Implemented

Allowed:

- Letters
- Numbers
- Spaces
- Basic punctuation

Blocked:

```
@ # $ % ^ & * { } [ ] < >
```

Benefits

- Spam prevention
 - Improved comment quality
 - Reduced malicious input
-

Part E – Multilingual Translation System

Requirement

Users should translate comments into their desired language.

Implementation

Implemented using:

```
translate-google
```

Supported Languages

- English
- Hindi
- French
- Spanish
- German

Workflow

1. User selects target language.
2. Clicks Translate.
3. Comment is translated.

4. Translation displayed below original comment.

Example

Original:

नमस्ते दोस्तों

Translated:

Hello Friends

Benefits

- Accessibility
- Global audience support
- Better user understanding

9. Database Design Enhancements

User Schema Enhancements

Added:

```
phone
state
city
otp
otpExpires
otpAttempts
lastOtpRequest
isVerified
plan
downloads
```

Comment Schema Enhancements

Added:

```
city
likes
dislikes
likedBy
dislikedBy
```

10. Security Enhancements

The following security features were implemented:

Authentication Security

- Firebase Authentication
- OTP Verification
- Email Validation

OTP Security

- Expiry Timer
- Cooldown Period
- Attempt Limitation

Comment Security

- Special Character Filtering
- Duplicate Vote Prevention
- Automatic Moderation

Database Security

- Unique Email Constraint
 - Validation Checks
 - Error Handling
-

11. Challenges Faced

Theme Synchronization

Issue: Some components ignored theme changes.

Solution: Replaced hardcoded Tailwind colors with dynamic theme variables.

Duplicate User Creation

Issue: Multiple records created during authentication.

Solution: Added unique email constraint and improved login flow.

Translation API Rate Limits

Issue: LibreTranslate public server restrictions.

Solution: Integrated translate-google package for reliable translation.

Comment Voting Abuse

Issue: Single user could vote repeatedly.

Solution: Implemented likedBy and dislikedBy tracking.

12. Conclusion

The project successfully implemented all required internship tasks while incorporating additional security, moderation, and personalization features.

The final platform now provides:

- Secure authentication
- Region-based OTP verification
- Dynamic theme adaptation
- Advanced comment moderation
- Multilingual comment translation
- User personalization
- Enhanced user engagement

The developed solution is scalable, secure, and suitable for real-world deployment with minimal additional configuration.

13. Future Enhancements

- Real SMS Gateway Integration
- AI-Based Comment Moderation
- Real-Time Notifications
- Multi-Factor Authentication
- Recommendation System

- Voice-Based Search
 - Advanced Analytics Dashboard
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Project Status

All Internship Requirements Successfully Implemented and Tested.